Severe Acute Malnutrition

136

Introduction

- ☑ Malnutrition is a silent killer that is underreported, underaddressed, and as a result under prioritized.
- ☑ It continues to be major cause of morbidity and mortality in children especially 6 months to 5 years and is mainly because of deficiency or imbalance intake of macronutrient (carbohydrate, protein, and fat) or micronutrient (vitamins, minerals, and trace element).
- ☑ The spectrum varies from underweight, stunted, moderately acute malnutrition, and severe acute malnutrition (SAM).
- ☑ Severe acute malnutrition is a medical and social problem as it accounts 6.4% of children below 60 months.
- ☑ However current estimated total population of India as 1,390 million, it is expected that about 8.1 million are likely to be suffering from SAM.

Status of Malnutrition in India

As per the global hunger index and the National Family Health Survey (NFHS)-5 (2019–2021):

- ☑ India is on place 67 among the 80 nations having the worst hunger situation
- ☑ 25% of all hungry people worldwide live in India
- ☑ 7.7% children below 5 years are severely wasted
- ☑ 35.5% are stunted and 19.3% are wasted (under 5 years) in the country

Factors Related to Malnutrition

It may be because of biological socioeconomic factors.

Socioeconomic factors
☑ Poverty
☑ Ignorance
☑ Gender female
☑ Rural
☑ Illiterate mother
☑ Low birth weight
Biological factors
☑ Maternal malnutrition and prematurity:
☑ Birth spacing < 3 years
☑ Age of mother 18–23 years
☑ Birth order > 3
☑ Underweight of mother
☑ Infectious diseases in mother
Environmental
☑ Unhygienic environment
☑ Drought and flood
☑ Wars and migration

WHO Classification of Malnutrition

Standard deviation (SD) score	***Height/length for age***	***Weight for age***	***Weight for height/length***
0 (median) to –2 SD	Normal	Normal	Normal
<–2 SD to –3 SD	Stunted	Underweight	Wasted or moderately acute malnutrition
<–3 SD	Severely stunted	Severely under-weight	Severely wasted or severe acute malnutrition

Criteria for diagnosing severely wasted or severely acute malnutrition: AS per the WHO and United Nations Children's Fund (UNICEF) the SAM is labeled if:

Parameters	6 months to 5 years	<6 months
Weight for height	<3 SD	<3 SD
Mid-upper arm circumference (MUAC)	<11.5 cm	Not included
Edema of both feet	Present	Present
Visible severe wasting	Not applicable	Present

WHO Classification of Malnutrition

- ☑ Visible severe wasting in children whose length is <45 cm.
- ☑ No other cause of edema, e.g., nephrotic syndrome or CHF, etc.
- ☑ For anthropometric measurements, a digital weighing machine with a sensitivity of 5 g, a wooden infant-cum-stadiometer and nonstretchable mid-upper arm circumference (MUAC) tapes can be used.

Criteria for Moderate Acute Malnutrition

- ☑ Weight for height <–2 SD to <–3 SD
- ☑ Mid-upper arm circumference (MUAC) 11.5–12.4 cm
- ☑ No edema

Criteria for Hospital Admission

- ☑ Weight for height/length <–3 SD/Z score of median of WHO child growth standard
- ☑ Bipedal edema
- ☑ If weight for height or weight for length cannot be measured, use the clinical finding of visible severe wasting

The effective management of SAM is done by taking the detail history, conducting thorough examination and laboratory tests, and protocol-based treatment.

History

- ☑ Recent intake of food and fluid and breastfeeding
- ☑ Diarrhea, vomiting, and appetite
- ☑ Loss of appetite
- ☑ Family circumstances
- ☑ Chronic cough, tuberculosis (TB), hemoptysis, and human immunodeficiency virus (HIV)
- ☑ Breastfeeding

Examination

- ☑ Anthropometry
- ☑ Pulse
- ☑ Respiratory rate
- ☑ Temperature
- ☑ Dehydration
- ☑ Capillary filling time/capillary refill time (CFT/CRT)
- ☑ Eye signs for vitamin a deficiency
- ☑ Fever
- ☑ Hypothermia
- ☑ Mouth ulcer
- ☑ Skin changes

Investigations

- ☑ Hemogram
- ☑ Renal function
- ☑ Serum electrolytes
- ☑ Screening for infection
- ☑ Urine routine microscopy (R/M)
- ☑ Urine culture
- ☑ X-ray chest

Level of Care of Children with SAM

It is decided on the basis of appetite test results (to be done in children aged >6 months) and associated medical complication as per following algorithm:

SAM

Check Medical Complication		
Yes	***No***	
Admit in NRC	Conduct appetite test	
	Pass	Fail
	Enroll in CSAM program	Admit in NRC

Phases of Management of SAM

Initial treatment	
Life-threatening problems identified	
	Specific deficiencies Metabolic abnormalities corrected
	Feeding begun
Rehabilitation	Intensive feeding
Emotional and physical stimulation	
Mother trained	

Follow-up

- ☑ Prevention of relapse
- ☑ Assured continued development

Ten Steps for Management of Children with SAM as per the WHO Recommendation

Since children with SAM need immediate attention because of increase chances of infection and increased mortality due to common childhood illness, especially hypoglycemia, hypothermia, and electrolyte imbalance, so the WHO has recommended 10 steps of management of children during stabilization and rehabilitation which are as follows:

S. No.	Steps	Stabilization phase		Rehabilitation phase
		Days 1–2	Days 3–7	Weeks 2–6
1.	Treat/prevent hypoglycemia	→	→	
2.	Treat/prevent hypothermia	→		
3.	Treat/prevent dehydration	→		
4.	Correct electrolyte imbalance	→	→	→
5.	Treat/prevent infection	→	→	
6.	Correct micronutrient deficiencies Iron supplementation	→	→	→
		No iron		Iron
7.	Start cautious feeding	→	→	
8.	Achieve catch-up growth			→
9.	Provide sensory stimulation and emotional support	→	→	→
10.	Prepare for follow-up			→

Step 1: Hypoglycemia

Criteria: Blood glucose < 54 mg/dL
Clinical feature: Poor feeding and lethargy convulsion

Management

- ☑ *If lethargic, unconscious, or convulsing:* Give 5 mL/kg 10% dextrose followed by 50 mL 10% dextrose or sucrose by nasogastric route. Repeat blood glucose after 30 minutes. If blood glucose remain <54 mg/dL then repeat the bolus.
- ☑ *Conscious*: Encourage feed 2 hourly
- ☑ *Prevention*: Two hourly feed and night feed should continue

Step 2: Treat/Prevent Hypothermia

- ☑ *Criteria:* Axillary temperature <35°C or rectal temperature <35.5°C
- ☑ *Clinical feature:* Lethargy and poor feeding
- ☑ *Management:*
  - Cover the head by cap and limbs by socks and mitten
  - Kangaroo mother care
- ☑ *Prevention:*
  - Maintain room temperature of 25°C
  - Feed immediately and then regular interval
  - Cover the child including head, soles, and palm
  - Stop draughts in the room
  - Promptly change wet clothes or bedding

Step 3: Treat/Prevent Dehydration

- ☑ *Treatment of dehydration in the children with SAM without shock:*
  - Following amount of oral rehydration is given as per weight of the child:

How often oral rehydration solution (ORS) to be given?	*Amount to be given*
Every 30 minutes for first 2 hours	5 mL/kg
Alternate hours for up to 10 hours	5–10 mL/kg

  - Add 20 mmol/L of potassium in 1 L low osmolar ORS.
  - Amount offered should be based on willingness to drink and amount of ongoing losses in stool.
  - Starter diet F-75 is given in alternate hours (2, 4, and 6) with reduced osmolarity ORS (3, 5, and 7) until child is rehydrated.
  - Sign of improved hydration and overhydration to be checked 1/2 hours for 1–2 hours then hourly

Sign of improved hydration: Any of following three parameters	*Sign of overhydration*
No longer thirsty	Increase pulse rate by 15 and respiratory rate by 5
Less lethargic	Engorged jugular vein
Slowing of pulse and respiratory rate from previous high rate	Puffiness of eye
Skin turgor is less slow	
Appearance of tear	
Action: Continue ORS	Stop ORS

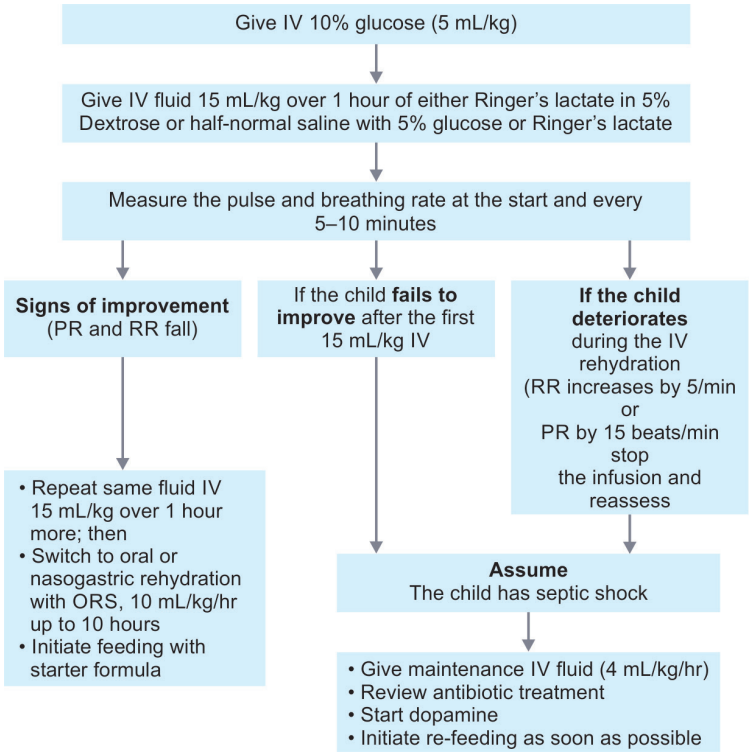
Step 3: Treat/Prevent Dehydration

Note: If diarrhea continues after rehydration give 50 mL or 100 mL ORS after each loose stool in <2 years or >2 years, respectively.

- Breastfeeding to continue with increased frequency.

☑ *Treatment of dehydration in the children with SAM having sign of shock and lethargic or has lost consciousness:*

- *Criteria of diagnosis:* Lethargic or unconscious and has cold hands along with either CRT >3 seconds or weak or fast pulse. It is managed through following algorithms:



Step 4: Correct Electrolyte Imbalance

☑ *Potassium chloride:* 3–4 mEq/kg/d for 14 days

☑ *Magnesium sulfate:*

- 0.3 mL/kg of maximum 2 mL intramuscular followed by 0.3 mL orally for 13 days
- Food without added salt in order to avoid sodium
- Do not treat edema with diuretics

All the children with SAM should receive empirical antibiotics as they are more prone to infection due to lower immunity and the type of antibiotics and duration depend upon the child with SAM with or without complication.

- ☑ *No complication in SAM child:* Oral amoxicillin 15 mg/kg 8 hourly for 5 days
- ☑ If child has complication then following protocols are used for treating the child with SAM:

Status	***Antibiotics***
All admitted cases with any complications other than shock, meningitis, or dysentery	☑ Injection ampicillin 200 mg/kg/day in 6 hourly and injection gentamicin 7 g mg OD for 7 days ☑ Injection cloxacillin 100 mg/kg/day 6 hourly if *Staphylococcus* is suspected
For septic shock or worsening/no improvement in initial hours	☑ Injection cefotaxime 150 mg/kg/day in three divided dose or ceftriaxone 100 mg/kg BD along plus injection gentamicin 7.5 mg OD
Meningitis	☑ Injection cefotaxime 50 mg/kg 6 hourly or ceftriaxone 100 mg/kg BD along plus injection amikacin 15 mg/kg in three divided dose
Dysentery	☑ Ciprofloxacin 15 mg/kg BD for 3 days. If already received or sick then injection ceftriaxone once day or divided in two doses for 5 days

Duration of Antibiotics

- ☑ *Suspicion of clinical sepsis:* At least for 7 days
- ☑ Urinary tract infections (UTI) for 7–10 days
- ☑ *Culture positive sepsis:* 10–14 days
- ☑ *Meningitis:* 14–21 days
- ☑ Deep seated infection like arthritis or osteomyelitis atleast for 4 weeks

Step 6: Correct Micronutrient Deficiencies and Iron Supplementation

Micronutrient supplementation is essential component of effective management of children with SAM. Following micronutrients are supplemented soon on admission and given for 2 weeks. However, iron in the dose of 3 mg/kg is started after 7 days of admission.

- ☑ *Vitamin A*: Vitamin A is given in the following dose according to age of the child:

Age	*Vitamin A dose*
<6 months	50,000 IU
6–12 months or if weight <8 kg	100,000 IU
<6 months	50,000 IU

- ☑ *Multivitamin supplement* (should contain vitamin A, C, D, E, and B_{12} and not just vitamin B complex): Twice recommended daily allowance.
- ☑ *Folic acid:* 5 mg stat then 1 mg for 13 days
- ☑ *Zinc*: 2 mg/kg for 14 days
- ☑ *Copper:* 0.3 mg/kg
- ☑ *Iron:* 3 mg/kg/day to be started after 7 days of patient admission
- ☑ Cautious feeding feeding

Step 7: Start Cautious Feeding

- ☑ In this phase, child with SAM is not able to tolerate usual amount of protein, sodium, and high amount of fat though recovers normal metabolic function and nutrition electrolytic balance without weight gain.
- ☑ Starter feed, i.e., F-75 started as soon 2–3 hourly and leftover feed recorded.

Step 8: Achieve Catch-up Growth

In order to achieve catch-up growth, catch-up diet, i.e., F-100 (calories 100, protein 2.9 g/100 mL started 3 hourly). It is started for weight gain, functional and immunological recovery.

If F-75 and F-100 are not available, nutritionists should calculate the amount of starter feed (prepared with cow's milk, sugar, rice powder, and vegetable oil to provide approximately 75 kcal and 0.9 g protein per 100 mL) and catch-up diet (to provide approximately 100 kcal and 2.9 g protein per 100 mL) to be offered to admitted SAM children, based on their daily weight measured between 10 AM and 11 AM.

Step 9: Provide Sensory Stimulation and Emotional Support

It reduces the risk of permanent mental retardation and emotional impairment. Physical and emotional stimulation are done by structured play therapy, encouraging child to laugh and vocalize.

Step 10: Prepare for Follow-up

- ☑ All SAM children should be followed up by health providers till s/he reaches weight-for-height of –1 SD. The frequency of follow-up is as follows:
  - 7 days after discharge
  - Fortnightly in first month, then
  - Monthly thereafter until WHZ reaches –1 SD or above. If problem is found, visit should be more frequent until it is resolved.
- ☑ *Measures to be taken*:
  - Examine
  - Weight record, height/length/MUAC
  - Feeding practices
  - Mental and physical stimulation
- ☑ *Criteria for discharge*:
  - Weight gain > 15%
  - Satisfactory weight gain > 5 g/kg/day for 3 consecutive days
  - Edema has resolved
  - Child eating an adequate amount of nutritious food that mother
  - Can prepare at home
  - All infections and other medical complications have been treated
  - Child is provided micronutrient
  - Up-to-date immunization

Step 10: Prepare for Follow-up

- ☑ *Discharge instruction*:
  - Continue any needed medications, vitamins, and iron
- ☑ *Danger signs*:
  - Not able to drink or breastfeed
  - Stop feeding
  - Develops fever
  - Fast or difficult breathing
  - Convulsion
  - Diarrhea and dysentery
- ☑ *Failure criteria of SAM management*:
  - Failure to gain appetite—day 4
  - Failure to start to lose edema—day 4
  - Edema still present—day 10
  - Failure to gain at least 5 g/day for 3 successive days
  - After feeding freely on catch-up diet
- ☑ Look for the cause of failure:
  - Insufficient food given
    - Adjustment of feeding problem
    - Corrected feed given
    - Correct amount offered at required time
    - Being fed adequately at night
    - Child encouraged to eat
    - Leftovers recorded
  - Vitamin or mineral deficiency
  - Unrecognized infections
  - Rumination

Carry Home Message

- ☑ SAM is major burden in developing country.
- ☑ SAM is a medical emergency.
- ☑ Pathophysiology is still elusive and incomplete.
- ☑ Ten steps are the key to successful management.
- ☑ Community-based treatment has revolutionized the management of SAM.
- ☑ Special needs for young infants and follow-up issues need to be recognized.

Further Reading

- ☑ Gardner G, Halweil B. Escaping Hunger, Escaping Excess. World Watch. 2000;13(4):24.
- ☑ Gardner G, Halweil Brian. Escaping Hunger, Escaping Excess. World Watch. 2000;13(4):5.
- ☑ Kumar P, Gupta P. Severe Acute Malnutrition. New Delhi: CBS Publishers & Distributors; 2017.
- ☑ Ministry of Health and Family Welfare. Operational Guidelines on Facility Based Management of Children with Severe Acute Malnutrition. New Delhi: Ministry of Health and Family Welfare; 2017.
- ☑ UNICEF. Improving Child Nutrition: The Achievable Imperative for Global Progress. New York: UNICEF; 2013.

Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Hypopigmentation

Lead Author
Sridhar Ganapathi

Co-Authors
Atanu Bhadra, Manish Gupta

Under the Auspices of the IAP Action Plan 2022

Remesh Kumar R
IAP President 2022

Upendra Kinjawadekar
IAP President-Elect 2022

Piyush Gupta
IAP President 2021

Vineet Saxena
IAP HSG 2022–2023

© Indian Academy of Pediatrics

IAP Standard Treatment Guidelines Committee

Chairperson

Remesh Kumar R

IAP Coordinator

Vineet Saxena

National Coordinators

SS Kamath, Vinod H Ratageri

Member Secretaries

Krishna Mohan R, Vishnu Mohan PT

Members

Santanu Deb, Surender Singh Bisht, Prashant Kariya, Narmada Ashok, Pawan Kalyan